PDF Extraction, Chunking, and Metadata

Building a clean and traceable document corpus

Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

1. PDF Is a Layout Format

A PDF preserves visual layout rather than guaranteeing a clean reading order. Text extraction can be affected by columns, tables, headers, footers, scanned pages, and unusual fonts. Some PDFs contain selectable text; others require optical character recognition. Always inspect extracted output before embedding it. A visually correct PDF can still produce scrambled text when processed automatically.

2. Extraction Quality Checks

For each file, record the file name, page count, extraction status, and any errors. Sample the first, middle, and last pages. Check whether headings remain connected to paragraphs, whether tables are understandable, and whether page breaks split sentences. If a page is scanned, OCR may be necessary and can introduce recognition errors. Poor extraction should be fixed before tuning embeddings or retrieval.

3. Chunking by Size or Structure

Fixed-size chunking is easy to implement and compare, but may split a definition or section in an awkward place. Sentence- or paragraph-aware chunking preserves more natural units, while heading-aware chunking can keep related material together. Overlap repeats some text between neighboring chunks to preserve boundary context. It can help, but too much overlap increases redundancy and storage.

4. Chunk Metadata

A chunk should retain enough information to identify its origin. Useful fields include document ID, source filename, page number, section heading, chunk index, and ingestion version. Metadata supports filters, citations, debugging, and reproducibility. Do not rely only on a chunk’s text to identify its source, since similar text may appear in multiple documents.

5. Choosing Chunk Parameters

There is no universally best chunk size. Small chunks can improve precision for narrow factual questions but may lack explanatory context. Larger chunks preserve context but can dilute relevance and consume more prompt tokens. Evaluate multiple settings with the same questions and embedding model. Track retrieval accuracy, answer groundedness, latency, and storage requirements.

6. A Safe Ingestion Workflow

Validate file type and size, extract text, normalize it conservatively, split it, reject empty chunks, create embeddings, and store vectors with metadata. Record counts at each stage: pages extracted, chunks generated, embeddings created, and records persisted. If counts unexpectedly differ, stop and investigate. Make ingestion idempotent where possible so uploading the same file does not silently create duplicates.

Key Takeaways

PDF extraction, chunking, and metadata determine the quality and traceability of the corpus. Inspect the data and measure alternatives before freezing the ingestion configuration.

End of document • Use source passages as evidence and verify claims during evaluation.